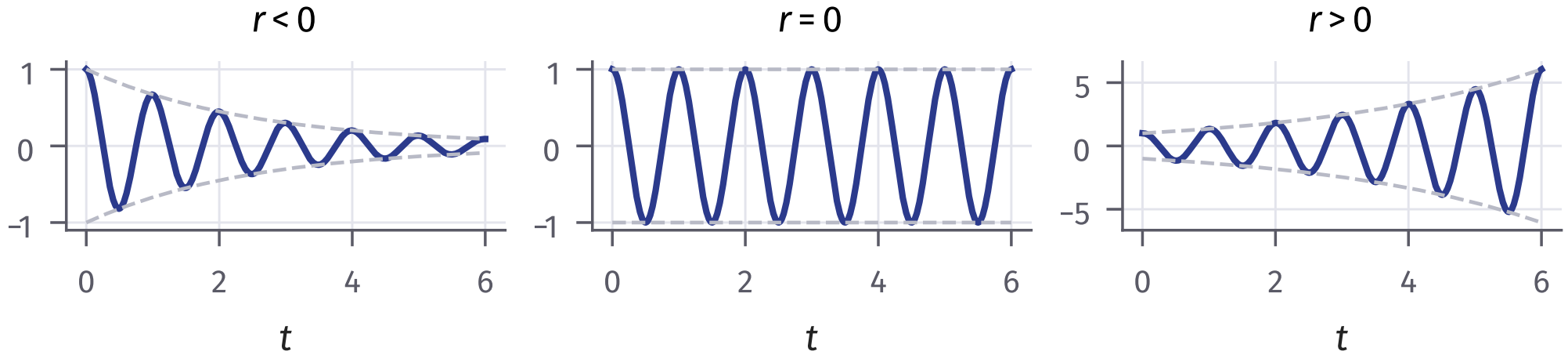$$\mathrm{Re}\{Ce^{at}\} = |C|e^{rt}\cos(\omega t + \phi)$$